

Deepak Vaid

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Publication List: [INSPIRE-HEP Database](#)

Personal

Citizen of India

Born on March 30, 1979

Education

Ph.D. Physics, Pennsylvania State University, USA 2003–2012

B.S. Physics, Magna Cum Laude, Missouri University of Science and Technology, USA¹, 2000–2003

Employment

Assistant Professor, National Institute of Technology Karnataka (NITK), July 2013–Present

Teaching Assistant, Pennsylvania State University, 2003–2007

Physics Tutor, University of Missouri at Rolla, 2000–2002

Research Highlights

A list of my papers can be found at <https://inspirehep.net/search?p=exactauthor%3AD.Vaid.1>

My work is on the interface of condensed matter and quantum gravity.

1. I have co-authored a book on Loop Quantum Gravity, which has recently been published by Springer (ref: [LQG for the Bewildered](#), also available online at [arXiv:1402.3586](#)).
2. In [arXiv:1607.00955](#) I have shown that Kepler's Second Law can be seen as a direct consequence of applying the *thermal time hypothesis* to the motion of planetary bodies.
3. In [arXiv:1312.7119](#) I proposed a relationship between Zhang's SO(5) theory (which unifies superconductivity and anti-ferromagnetism) with gravity arising from the so-called BF formulation. I've conjectured that different spacetimes can be thought of as different phases of an underlying condensed matter theory.

¹[previously known as](#) "University of Missouri at Rolla" (UMR)

4. In [arXiv:1208.3335](#) I pointed out the relationship between the large diffeomorphisms of black hole horizons and the physics of the quantum hall effect. This relationship was also elaborated on by other authors in [arXiv:1402.2274](#). In the same paper I proposed that the dynamics of a black hole horizon can be modeled using the Hubbard model. This idea has been rediscovered recent in work by other authors [arXiv:1703.01156](#)
5. In [arXiv:1307.0096](#) I proposed a relationship between elementary particles and quantum computational gates. This model provides a concrete basis for the so-called computational universe hypothesis.
6. The origin of gauge fields corresponding to the strong, weak and electromagnetic forces remains a mystery. Several proposals exist:
 - (a) Grand Unified Theories (GUTs) based on symmetry breaking of large gauge groups ($SO(5)$ or $SO(10)$),
 - (b) String models where Standard Model gauge fields and various scalar fields arise due to compactification of the 10D stringy spacetime,
 - (c) As additional degrees of freedom which can be attached to spin-networks in the LQG approach to quantum gravity.

In [arXiv:1309.0652](#) I have proposed that, instead, that so-called "fundamental" gauge fields might instead correspond to emergent degrees of freedom required in order to explain the interaction between defects in spin-networks which provide the microscopic substructure for spacetime in LQG.

Publications

1. [LQG for the Bewildered](#), (with S. Bilson-Thompson), Springer 2017, [arXiv:1402.3586](#),
2. Thermal Time and Kepler's Second Law, 2016, [arXiv:1607.00955](#)
3. Quantum Gravity for Dummies, 2014, [arXiv:1402.2757](#)
4. Superconducting and Anti-Ferromagnetic Phases of Spacetime, 2013, [arXiv:1312.7119](#)
5. Non-abelian Gauge Fields from Defects in Spin-Networks, 2013, [arXiv:1309.0652](#)
6. Elementary Particles as Gates for Universal Quantum Computation, 2013, [arXiv:1307.0096](#)
7. Quantum Hall Effect and Black Hole Entropy in Loop Quantum Gravity, 2012, [arXiv:1208.3335](#)
8. Embedding the Bilson-Thompson model in an LQG-like framework, 2010, [arXiv:1002.1462](#)
9. Local Void vs Dark Energy: Confrontation with WMAP and Type Ia Supernovae (with S. Alexander, T. Biswas and A. Notari), 2008, [arXiv:0712.0370](#)
10. A fine tuning free resolution of the cosmological constant problem (with S. Alexander), 2007, [arXiv:hep-th/0702064](#)
11. Gravity Induced Chiral Condensate Formation and the Cosmological Constant (with. S. Alexander), 2006, [arXiv:hep-th/0609066](#)

Talks, Seminars, Research Visits

1. Research interaction with Prof. Romesh Kaul and Prof. Ghanshyam Date at IMSc Chennai from Apr 5-11, 2017
2. *Phases of Spacetime: Gravity as a Condensate of Gauge Fields*, Poster presentation and “Promotional Talk for Young Scientist”, DAE-BRNS Symposium on High Energy Physics at Delhi University, December 12-16, 2016 (ref: [arXiv:1312.7119](#))
3. Research Interaction undertaken under TEQIP-II program of Govt of India with Prof Pablo Arrighi at ENS Lyon, France & LIF Marseille, France, June 9-20, 2016
4. *Phases of Spacetime: Gravity as a Condensate of Gauge Fields*, Poster presentation, International Conference on Gravitation and Cosmology (ICGC 2015) at IISER Mohali, India, December 14-18, 2015 (ref: [arXiv:1312.7119](#))
5. *Phases of Spacetime: Gravity as a Condensate of Gauge Fields*, Seminar, School of Physics, University of Hyderabad, Hyderabad, India, September 7, 2015 (ref: [arXiv:1312.7119](#))
6. *The ABC of Quantum Gravity*, National Institute of Technology Karnataka (NITK), INSPIRE Talk, Surathkal, India, December 16, 2013 (ref: [arXiv:1402.2757](#))
7. *Loop Quantum Gravity for the Bewildered*, Colloquim, Center for Theoretical Physics, Jamia Milia Islamia, New Delhi, India, June 14, 2012 (ref: [arXiv:1402.3586](#))
8. *Loop Quantum Gravity for the Bewildered*, Three Seminars given at Physics Department, University of Adelaide, Adelaide, Australia, August 19, 22 and 29, 2011 (ref: [arXiv:1402.3586](#))
9. *Elementary Particles as Gates for Universal Quantum Computation*, Colloquim at Center for High Energy Physics, Indian Institute of Science, Bangalore, India, April 21, 2010 (ref: [arXiv:1307.0096](#))
10. *Elementary Particles as Gates for Universal Quantum Computation*, Seminar at Mehta Research Institute, Allahabad, India, April 7, 2010 (ref: [arXiv:1307.0096](#))
11. October 2009, [Visiting Researcher](#) at Perimeter Institute for Theoretical Physics, Waterloo, Canada

Teaching Activities

Assistant Professor, National Institute of Technology Karnataka (NITK):

Modern Physics, Introductory Modern Physics Course for B. Tech 1st year students, Aug 2013 – Present (Seven semesters)

Instructor, Aug 2013 – Present (Six semesters)

Course Co-ordinator, Aug 2014 – May 2015

Instructor, *Mathematical Physics*, M.Sc. Physics 1st year, Aug 2013 – Dec 2013

Instructor, *General Relativity*, M.Sc. Physics Elective, Jan 2014 – May 2014

Instructor, *Nuclear Physics*, M.Sc. Physics 2nd year, Aug 2014 – Dec 2014, Aug 2016 – Dec 2016

Instructor, *Quantum Hall Effect*, as part of *Material Synthesis & Characterization*, Oct 2015

Instructor, *Superconductivity*, as part of *Condensed Matter Physics, Part 2*, Oct 2016 – Nov 2016

Teaching Assistant, Pennsylvania State University:

Conducted Labs for Introductory Physics Courses, Aug 2003 - Dec 2007 (Seven Semesters + Four Summer Sessions)

Last updated: August 30, 2017

Teaching Experience (Theory)				
Course Code	Course Title	PG/UG	Period	Credit Hours
PH110	Physics (*)	UG	Jul-Dec, 2013	4
PH701	Mathematical Physics	PG	Jul-Dec, 2013	4
PH110	Physics	UG	Jan-May, 2014	4
PH876	General Relativity	PG	Jan-May, 2014	3
PH110	Physics	UG	Jul-Dec, 2014	4
PH862	Nuclear Physics	PG	Jul-Dec, 2014	2 (**)
PH110	Physics	UG	Jan-May, 2015	4
PH110	Physics	UG	Jul-Dec, 2015	4
PH914	Applied Quantum Mechanics	PG	Jul-Dec, 2015	3
PH879	Physics of Low Dimensional Systems	PG	Jul-Dec, 2015	1 (***)
PH110	Physics	UG	Jan-May, 2016	4
PH110	Physics	UG	Jul-Dec, 2016	4
PH802	Nuclear Physics	PG	Jul-Dec, 2016	2 (**)
PH801	Condensed Matter Physics, Part II	UG	Jul-Dec, 2016	1 (****)
PH110	Physics	UG	Jan-May, 2017	4
PH753	Statistical and Thermal Physics	PG	Jan-May, 2017	2 (**)
PH876	General Relativity	PG	Jul-Dec, 2017	3

Table 1: *Legend:* (*) 1st year introductory modern physics course for B.Tech students; (**) 4 credit hour course shared with Prof H. S. Nagaraja; (***) Taught “Hall Effect” module as part of PH879 (overall course instructor Prof Partha P. Das); (****) Taught “Superconductivity” module as part of PH801 (overall course instructor Prof Kasturi Bangera)

Computational Skills

Proficient in: Python, Jupyter (previously IPython Notebook), C++, $\text{\LaTeX}$, SAGEMath, Mathematica, R, Matlab

Platforms: Apple OS X, Ubuntu Linux, Microsoft Windows

References

[Bindu Bambah](#), Professor, School of Physics, University of Hyderabad, Hyderabad - 500046, Telangana, India, +91-9849976697, +91-40-23011403, bbsp@uohyd.ernet.in

[G. Umesh](#), Professor, Department of Physics, National Institute of Technology Karnataka, Surathkal - 575025, Karnataka, India, +91-9901184376, +91-824-2473276, umesh52@gmail.com

[Martin Bojowald](#), Associate Professor of Physics, 304A Whitmore Lab, Pennsylvania State University, PA, USA, +1 814 865 3502, bojowald@gravity.psu.edu

[Jainendra K. Jain](#), Erwin W. Mueller Professor of Physics, 104 Davey Laboratory Rm. 207, Pennsylvania State University, PA, USA, +1 814 574 7497, jain@phys.psu.edu

Research Supervision

Ph.d. Students Supervision

Name	Title	From	To
Ahmed Rizwan C. L.	Holographic Superconductivity in Loop Quantum Gravity	Dec, 2013	Present
Rajani K. V.	Quantum Hall Effect in Loop Quantum Gravity	Dec, 2014	Present
Safir T. K.	Holographic Quantum Computation and Quantum Error Correction	Dec 2014	Present

M.Sc. Students Supervision

Name	Title	From	To
Merin Joseph	Chaotic Dynamics in Complex Systems	Aug, 2016	May, 2017
Shwetha B.	Relativistic Velocity Addition Law Without the Second Postulate	Aug, 2015	May 2016
Sreedevi Varma N.	Alcubierre's Drive: A Study Using ADM Formalism and Energy Conditions	Aug, 2015	May, 2016
Priyamedha S K R	Electrodynamics of Cavity Resonators	Aug, 2014	May, 2015
Sarath E.	Perturbation Theory and Relativistic Corrections	Aug, 2014	May, 2015

Research Internships

Name	Home Institute	Project Title	Period
Vinayaka Garg	Vellore Institute of Technology, Chennai	1D Ising Model	Dec 2015
Neel Kumar	Vellore Institute of Technology, Chennai	Simulating Planetary Orbital Motion	Jun - Jul, 2016
Karthik Suresh	Central University of Karnataka, Gulbarga	Conductance in Mesoscopic Systems via Non-equilibrium Green's Functions	Jun - Jul, 2017
Shalmali Sudhindra	Alva's College, Moodbidri	Non Equilibrium Greens function and Electrical conductivity in small scale systems	Jun - Jul, 2017